



Exploring factors influencing teacher self-efficacy in implementing inclusive education in Cambodia: A two-level hierarchical linear model

Sokunrith Pov^{a,*}, Norimune Kawai^b, Rie Murakami^c

^a Graduate School of Humanities and Social Sciences, Hiroshima University, Japan

^b The Institute for Diversity and Inclusion, Hiroshima University, Japan

^c Graduate School of Education, Hiroshima University, Japan

ARTICLE INFO

Keywords:

Teachers' self-efficacy
Inclusive education
Inclusive practice
Students with disabilities
Hierarchical linear model
Cambodia

ABSTRACT

This study examined teachers' self-efficacy for inclusive practices and identified factors influencing it. The study involved 1008 primary school teachers from 236 schools across Cambodia. It was guided by a conceptual framework grounded in Bandura's Theory of Self-Efficacy, addressing existing research gaps and incorporating empirical insights into the factors shaping teachers' self-efficacy for inclusive practices. Using descriptive statistics, exploratory factor analysis, reliability tests, and the two-level hierarchical linear model (HLM), the study found that teachers reported the highest self-efficacy in managing behaviours and the lowest in collaboration, especially with parents. Factors such as gender, interaction experience, inclusive education training, knowledge of disabilities, and the implementation of inclusive education programmes significantly influenced their self-efficacy. The study's findings provide valuable insights for improving teacher education in the context of inclusive practices.

1. Introduction

There has been a significant shift in international education movements catering for students with disabilities (SWDs). Initially, it stems from the provision of special education, which has then transitioned to integrated education, where SWDs are included but often remain segregated within regular schools. The current trend strongly advocates for inclusive education (IE), highlighting the importance of ensuring that all students, including those with special educational needs and disabilities (SEND), are fully included and supported alongside their peers in regular education settings (Kuroda et al., 2017).

In Cambodia, the Law on Education assures the right of SWDs to be educated in regular schools alongside their peers without disabilities and receive additional support from teachers, as outlined in Article 39 (Cambodia National Assembly, 2007). This commitment aligns with the Salamanca Statement and Framework for Action, asserting the right to education of every individual within regular education settings. It further emphasises the right of children with SEND to receive additional support and encourages the governments to adopt the principle of IE into their education system (United Nations Educational, Scientific, and

Cultural Organisation (UNESCO, 1994).

In 2008, the Cambodian Ministry of Education, Youth and Sport (MoEYS) implemented the Policy on Education for Children with Disabilities (PECD) to promote the integration of SWDs into the regular education system (MoEYS, 2008). In 2018, this policy was revised into the Policy on IE (PIE), aiming to ensure all individuals with SEND 'have rights to inclusive and equitable quality education and opportunities for life-long learning' (MoEYS, 2018, p. 4), which aligns with the vision of the Sustainable Development Goal 4 on education (UNESCO, 2015).

It has been observed that generally, regular schools in Cambodia admit students with mild disabilities (Kalyanpur, 2016). A small number of schools are more inclusive, accommodating students with a broader range of conditions and disabilities, including hearing impairments, visual impairments, and intellectual disabilities (Ravet & Mtika, 2021). These schools often collaborate with development partners and non-governmental organisations specialising in disability support. According to Kalyanpur (2016) and data from the Education Management Information System (EMIS) of MoEYS in the academic year 2021–2022, disabilities in regular schools are categorised into four main types: physical, mental, intellectual, and other disabilities. Notably, physical

* Correspondence to: Graduate School of Humanities and Social Sciences, Hiroshima University, 1-5-1 Kagamiyama, Higashi-Hiroshima, Hiroshima 739-8529, Japan.

E-mail address: sokunrithp@gmail.com (S. Pov).

¹ <https://orcid.org/0000-0002-2595-2396>

<https://doi.org/10.1016/j.stueduc.2024.101415>

Received 16 November 2023; Received in revised form 16 August 2024; Accepted 19 October 2024

Available online 30 October 2024

0191-491X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

disabilities comprise movement disorders, visual impairments, hearing impairments, speech impairments, and internal disorders or chronic diseases (see MoEYS, 2023).

Despite various commitments of the government and MoEYS, SWDs encounter significant challenges in regular schools across Cambodia. The most recent general population census of the Ministry of Planning (MoP) in 2019 reported that the total population was 15,552,211. Approximately 4.9 % of the population had disabilities, ranging from mild to severe conditions. Among individuals aged seven years and older with disabilities, about 22.9 % attended primary schools, while 55.7 % did not complete primary education. Only 17 % reached lower secondary education, 1.4 % completed upper secondary education (diploma), and 1.3 % obtained higher education qualifications (MoP, 2020).

Data from EMIS of MoEYS indicated that, in the academic year 2021–2022, the primary education level had the highest number of SWDs, with 21,335 students, compared to 779 in kindergarten, 5505 in lower secondary level, and 4134 in upper secondary level (MoEYS, 2023). The prevalence of SWDs was particularly high at the primary level, posing a significant hurdle for Cambodia in successfully developing an IE system (Disability Action Council, 2003).

In Cambodia, SWDs remain encountering a myriad of formidable obstacles when it comes to accessing and successfully navigating the educational cycles. A significant number of SWDs in Cambodia drop out of schools before completing a given academic phase, primarily due to an array of persistent barriers stemming from structural, attitudinal, and environmental factors that remain unaddressed (see Ravet & Mtika, 2021). Consequently, the majority of SWDs find themselves unjustly excluded or marginalised within regular educational settings, often under the pretext of an ‘inclusive class’ (Nishio, 2019). Their needs extend beyond mere school attendance and encompass the imperative requirements of appropriate provisions and tailored assistance from teachers, enabling their active participation within regular schools (Disability Action Council, 2003). The principle of inclusion extends beyond the presence of students in regular schools. It necessitates changes to values, attitudes, policies, and especially teachers’ practices to ensure full inclusion (Polat, 2011).

Teachers play a crucial role in implementing inclusive practices and ultimately determining the success or failure of IE (Forlin & Lian, 2008). They translate IE policies into practical changes in the classroom, making it imperative to address their self-efficacy in implementing IE (Shah et al., 2016). Understanding teachers’ self-efficacy (TSE) for inclusive practices is imperative to making schools inclusive for everyone (Wray et al., 2022) and informing the changes in teacher preparation programmes and policies (Sharma et al., 2012).

A large body of literature highlights the pivotal role of TSE in determining the successful implementation of IE across various contexts (e.g., Alnahdi, 2019; Malinen et al., 2013; Monteiro et al., 2019; Ravet & Mtika, 2021; Sharma et al., 2012; Vogiatzi et al., 2022; Wray et al., 2022; Yada & Savolainen, 2019). Research has consistently found that TSE is the most powerful predictor influencing their attitudes, behavioural intentions towards inclusion, and behaviour to include students with SEND in regular education settings (e.g., Emmers et al., 2020; Urton et al., 2023; Weisel & Dror, 2006; Yan & Sin, 2014). Moreover, the investigation of factors influencing TSE holds significant insights into addressing barriers to the development of IE. It highlights how teachers should be supported and well-equipped to work in inclusive classroom settings (Sharma et al., 2012; Vogiatzi et al., 2022).

While research emphasises various fundamental aspects of understanding TSE and its influential factors, there is a noticeable dearth of studies exploring these at the school level in Cambodia. Previous studies have primarily focused on project evaluations and policy reviews (e.g., Kalyanpur, 2011; Kalyanpur, 2014; Kalyanpur, 2016; Kuroda et al., 2017; Nishio, 2019), often neglecting to explore TSE and the barriers hindering their self-efficacy beliefs in implementing IE in regular classroom settings. This gap underscores the need to understand Cambodian teachers’ perceived self-efficacy in areas such as

collaborating with colleagues and professionals, using inclusive instruction, and managing the behaviours of diverse students (see Sharma et al., 2012), particularly in the context of large class sizes and the high prevalence of SWDs at the primary level (see MoEYS, 2022b; MoEYS, 2023).

It is also critical to explore a wide range of factors that may influence Cambodian TSE in supporting SWDs within regular schools. Addressing these aspects is crucial for overcoming inclusion barriers, implementing effective interventions and practices at the school level, and driving policy and program changes that support in-service teachers’ inclusive practices and promote the full inclusion of SWDs in regular education settings (Mendoza & Heymann, 2022).

2. Aims and research questions

This study aimed to employ a nationwide survey to examine in-service TSE in implementing inclusive practices and the factors influencing it at regular primary schools in Cambodia. It addressed the following research questions:

1. What were the general and specific self-efficacy for inclusive practices of Cambodian teachers regarding managing students’ behaviours, collaboration, and using inclusive instruction?
2. What individual- and school-level factors significantly influenced Cambodian teachers’ general self-efficacy for inclusive practices?

3. Literature review

3.1. Teachers’ self-efficacy for inclusive practices

To successfully implement IE, teachers need support beyond having positive attitudes towards inclusion and a good understanding of the IE concept. They need guidance in applying these concepts within their specific teaching contexts. This support helps enhance their self-efficacy and confidence to implement inclusive practices in IE classrooms (Woodcock et al., 2022). This raises concerns about how teachers perceive their self-efficacy to implement inclusive practices, as most teacher preparation and continuous professional development programmes in many developing countries, including Cambodia, are not designed to effectively prepare teachers to work in inclusive classroom settings (Kuroda et al., 2017; Loreman et al., 2013).

To address this issue, several previous studies have proposed various hypothetical models to assess TSE and its influential factors (e.g., Alnahdi, 2019; Malinen et al., 2013; Yada & Alnahdi, 2021), grounded in Bandura’ Theory of Self-Efficacy (see Bandura, 1977). Bandura’s Theory of Self-Efficacy asserts that ‘expectations of personal efficacy determine whether coping behaviour will be initiated, how much effort will be expended, and how long it will be sustained in the face of obstacles and aversive experience’ (p.191). These expectations are shaped by four main sources of self-efficacy: performance accomplishments, vicarious experience, verbal persuasion, and physiological states (Bandura, 1977).

In the context of teaching in inclusive classrooms, research shows that these sources are critical factors influencing teachers’ self-efficacy in consistently implementing IE. Malinen et al. (2013) maintain that teachers’ vicarious experiences involve observing others’ teaching practices, such as teaching SWDs in regular schools. Verbal persuasion refers to interactions where teachers receive feedback on their abilities to effectively accomplish given tasks, often from colleagues, special education teachers, or therapists. Vicarious experiences and verbal persuasion can be developed and enhanced through collaborative efforts at the school and classroom levels, where teachers work with colleagues and other professionals to support SWDs (Horne & Timmons, 2009; Pijl, 2010; Valdés & Gómez-Hurtado, 2023).

Physiological states refer to feelings of anxiety or excitement that can be interpreted as indicators of competence or inability to handle specific

activities, such as teachers' concerns about inclusion and confidence in implementing inclusive practices. Performance accomplishments are conceptualised as teachers' mastery experiences, including both success and failure when implementing IE (Malinen et al., 2013). Success fosters mastery expectations, while repeated failures lower them. Once strong efficacy expectations are established through repeated successes, occasional failures are less likely to have a significant negative impact (Bandura, 1977). Research shows that successful teaching experiences with SWDs are the most powerful predictor of resilient self-efficacy, enabling teachers to persistently cope with challenges in implementing IE (Chao et al., 2016; Malinen et al., 2013; Monteiro et al., 2019). Overall, these sources collectively have a significant impact on TSE, teachers' behaviours in the classroom, and students' learning outcomes (Tschannen-Moran et al., 1998).

Drawing on Bandura's Theory of Self-Efficacy, Sharma et al. (2012) argue that self-efficacy is a key factor driving teachers to create inclusive learning environments and effectively implement inclusive practices, thereby fostering student learning outcomes. Teachers with high self-efficacy have positive attitudes about the benefits of inclusion and are more motivated to implement inclusive practices (Alnahdi, 2019; Sharma et al., 2012; Yada & Alnahdi, 2021). To measure TSE, specifically in the context of IE, Sharma et al. (2012) developed the Teacher Efficacy for Inclusive Practices (TEIP) scale. This scale measures TSE regarding managing behaviours, collaboration, and using inclusive instruction in supporting students with SEND. It emphasises the skill sets that teachers need to successfully implement inclusive practices, highlighting a social rather than a medical model of disability (Sharma et al., 2012).

3.2. Factors influencing teachers' self-efficacy

In addition to the importance of understanding self-efficacy and its sources, understanding the factors influencing self-efficacy in implementing IE provides practical insights for enhancing policies and practices that promote TSE and IE (Wray et al., 2022). These factors are related to the four sources of self-efficacy, shaping resilient teachers' confidence and self-efficacy beliefs in supporting SWDs. Research shows multifaceted findings regarding TSE for inclusive practices and its perceived barriers across both developing and developed countries (e.g., Chao et al., 2016; Malinen et al., 2013; Monteiro et al., 2019; Ravet & Mtika, 2021; Vogiatzi et al., 2022; Wray et al., 2022; Yada & Savolainen, 2019).

Malinen et al. (2013) utilised Bandura's Theory of Self-Efficacy to investigate TSE and the factors influencing it in China, Finland, and South Africa. The study revealed that in China, factors such as experience teaching SWDs, interaction with people with disabilities (PWDs), and teacher types significantly impacted all areas of Chinese TSE, including using inclusive instruction, collaboration, and managing student behaviour. In Finland, experience teaching SWDs, the amount of IE training (IET), and gender were the significant predictors of Finnish TSE. Similarly, the study found that in South Africa, experience teaching SWDs, interaction with PWDs, and age were the significant predictors.

Monteiro et al. (2019) conducted a mixed-method study examining TSE in implementing IE in Macao and identified various significant factors. The study found that teachers were not well-prepared to teach SWDs and lacked the necessary knowledge and skills to support SDWs. Increased knowledge, confidence, experience of IET, age, and being male were significantly associated with increased TSE, particularly in managing students' disruptive behaviour. Similarly, Chao et al. (2016) provided a week-long IET programme for in-service teachers in Hong Kong and measured factors influencing their self-efficacy before and after the training. The study found that knowledge of legislation and policies, confidence in teaching SWDs, and experience teaching SWDs significantly influenced TSE.

Kuyini et al. (2020) explored primary school TSE beliefs, attitudes, and concerns regarding implementing IE in Ghana. Despite having

positive attitudes towards inclusion, Ghanaian teachers expressed significant concerns and reported moderate self-efficacy in implementing IE. Gender, age, and years of teaching experience were not the significant factors influencing their self-efficacy. However, experience teaching in special education classrooms and experience teaching SWDs significantly impacted TSE. Teachers with more experience teaching SWDs reported a significantly higher sense of self-efficacy to implement IE than those with less such experience. Teaching experience with SWDs beyond special education classrooms significantly enhances general teachers' confidence in delivering inclusive instruction, working with professionals to support SWDs, and effectively managing the behaviour of all students (Malinen et al., 2013).

Kuyini et al. (2020) also found that the amount of IET was a strong predictor of TSE for implementing IE. Teachers with more training in teaching SWDs had significantly higher levels of self-efficacy compared to those without such training. Furthermore, research indicates that exposure to inclusive classroom settings and extended opportunities to teach SWDs significantly enhance TSE. Therefore, schools with IE programmes should ensure that all teachers have the opportunity to teach SWDs (Peebles & Mendaglio, 2014; Specht & Metsala, 2018).

In a broader context, Wray et al. (2022) provide crucial insights into factors influencing TSE for inclusive practices through a review of studies conducted over the past ten years. Wray et al. (2022) systematically reviewed 71 research articles from over 27 countries to identify common factors influencing TSE. They categorised these factors into five variable components: (1) demographics, (2) teacher education and professional learning, (3) school climate, (4) experience interacting with PWDs, and (5) teacher internal attributes.

Wray et al. (2022) identified six variables within demographics, including knowledge of IE and legislation, country and context (e.g., system and culture), teaching settings (e.g., primary or secondary schools), years of teaching experience, age, and gender. Teacher education and professional learning encompassed pre-service education, practicum experience, and in-service professional learning (e.g., experience of IET). School climate included four main aspects: leadership support for IE, collaboration, whole-school processes, and shared practices. Notably, no studies have examined the variability of TSE in different school locations, such as rural versus urban areas. Wray et al. (2022) also highlighted that teachers' experience interacting with PWDs in the community and experience teaching SWDs significantly influenced their self-efficacy. For teacher internal attributes, several variables were identified as significant predictors of TSE, including attitudes towards inclusion, beliefs about IE, confidence in their ability to implement IE, and personality traits (Wray et al., 2022).

To date, no studies have comprehensively analysed these factors within the five variable components in any analytical models. Malinen et al. (2013) argue that most previous studies have primarily examined the relationships between TSE, coping with behavioural issues, and attitudes towards inclusion, with less focus on the broader range of influential factors.

3.3. Conceptual framework

Fig. 1 presents the conceptual framework of the current study, which is grounded in Bandura's Theory of Self-Efficacy. It addressed existing research gaps, incorporated empirical insights into factors influencing

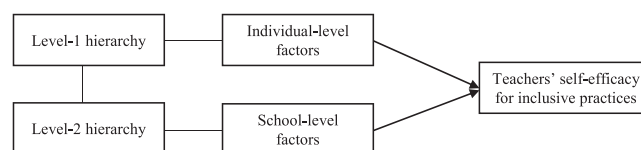


Fig. 1. Conceptual Framework of the Two-Level Hierarchical Linear Model Assessing Factors Influencing Teachers' Self-Efficacy for Inclusive Practices.

TSE, and utilized the two-level hierarchical linear model (HLM). Research shows that most previous studies have primarily utilised t-test, ANOVA, regression, and Structural Equation Modelling (SEM) to test their hypothetical models, typically incorporating only a few predictor variables such as gender, experience of IET, and experience teaching SWDs (e.g., Malinen et al., 2013; Monteiro et al., 2019; Yada & Savolainen, 2019). As a result, these studies provide a limited view of factors influencing TSE by not accounting for the complex data structure. While having a sound analytical model is important, evaluating a wide range of predictors related to individuals (teachers) and schools is imperative for informing policy and practice changes to enhance teachers' inclusive practices (Alnahdi, 2019).

To test a hypothetical model that encompasses a broad range of predictors at different levels, the current study employed the two-level HLM for data analysis. HLM has not been utilised to investigate TSE and its influential factors for inclusive practices. In principle, HLM is employed to assess the structure or nested data considered at various hierarchical levels (Matsuyama, 2020; Woltman et al., 2012). Historically, HLM has been utilised in educational research to address issues where hierarchies exist naturally (Sullivan et al., 1999), with nested data at multiple levels (e.g., teacher-level data and school-level data, with teachers nested within schools) (Garson, 2013).

In the current study, teachers-related variables were nested at Level 1 of the hierarchies, while school-related variables were nested at Level 2. The analysis included factors related to previous studies' findings and the five-factor components identified by Wray et al. (2022). As explained earlier, various factors have been found to significantly impact TSE for inclusive practices in different contexts. Additionally, variables from previous studies pertinent to the context of Cambodia were added in the models, as shown in Table 5.

Drawing on previous studies, particularly Wray et al. (2022), a total of 20 predictor variables were included in the analytical models of the current study. These variables involved gender, age, disability, experience interacting with PWDs, family or relatives with disabilities, teacher training programmes, experience with IET, sufficient IET, knowledge of PECD, knowledge of PIE, knowledge of IE, knowledge of special education, knowledge of disabilities, knowledge of Individualised Education Plans (IEPs), years of teaching experience, experience teaching SWDs, confidence in teaching SWDs, collaboration experience, schools implementing IE programmes, and school location. These variables were categorised and nested within different hierarchies of the two-level HLM. Of the 20 predictor variables, 18 were classified as 'individual-level factors' and were nested within Level 1 of the hierarchical structure. Furthermore, the remaining two variables were designated as 'school-level factors' and nested within Level 2 of the hierarchical structure.

4. Methods

4.1. Research design

This study employed a quantitative approach with a survey design to investigate Cambodian TSE for inclusive practices regarding managing students' behaviours, collaboration, and using inclusive instruction and factors influencing their self-efficacy. It included a large number of primary school teachers from all provinces and cities in Cambodia for the investigation. The survey design was chosen to investigate a broad range of teachers and schools from different locations and socioeconomic contexts across the country, ensuring a diverse representation. This approach enhances the validity of generalisation (Fowler, 2009) and allows the study to use the two-level HLM for data analysis. The two-level HLM requires a large number of participants and schools to generate accurate results.

4.2. Participants

The selection of schools and participants was based on (1) the school location, encompassing both urban and rural areas, and (2) the historical inclusion of SWDs in the schools. We acquired a list of all primary schools in Cambodia for the academic year 2021–2022 from the EMIS of MoEYS. The schools were then filtered based on location and the track record of SWDs. Subsequently, we categorised the selected schools into rural and urban areas and conducted a random selection of primary schools from each province. As a result, a total of 236 primary schools across 25 provinces and cities in Cambodia were randomly chosen as research sites. We informed and invited all teachers at the selected schools to participate in the study. As a result, a total of 1008 primary school teachers voluntarily participated and completed the questionnaire, with a response rate of 71 %.

Among the participants, 33 % were male, and 67 % were female. Regarding teaching experience, 27 % had one to five years of teaching experience, 22 % had six to ten years, and 51 % had 11 or more years of experience. Of the participants, 45 % had prior experience in teaching SWDs. Approximately 71 % were from rural schools, and the remaining 29 % were from urban schools. Notably, 50 % of the selected schools had implemented IE programmes, while the remaining 50 % had not.

4.3. Research instrument

This study employed a self-reported questionnaire for data collection, consisting of two sections: teachers' background information and the Cambodian version of the TEIP scale. The section on teachers' background information collected predictor variables related to their personal details, educational background, and professional experience. In line with the 'Conceptual Framework', this study designed the questionnaire that incorporated predictor variables identified in prior research. Specifically, it drew on predictors outlined in the review study by Wray et al. (2022), which encompassed five key factor components: (1) demographics, (2) teacher education and professional development, (3) school climate, (4) experience in interacting with PWDs, and (5) teacher internal attributes. In total, this study included 20 predictor variables for analysis in the two-level HLM.

Individual-level factors (Level 1) comprised 18 predictors, encompassing variables such as gender, age, disability, experience interacting with PWDs, family or relatives with disabilities, teacher training programmes, experience with IET, adequate IET, knowledge of PECD, knowledge of PIE, knowledge of IE, knowledge of special education, knowledge of disabilities, knowledge of IEPs, years of teaching experience, experience of teaching SWDs, confidence in teaching SWDs, and collaboration experience. School-level factors (Level 2) included whether schools were implementing IE programmes and their geographical location. Details for each variable are provided in Appendix 1.

The study adopted the TEIP scale from Sharma et al. (2012) to assess the levels of Cambodian TSE for inclusive practices. The TEIP consists of 18 items that are rated on a six-point scale (1 = strongly disagree; 2 = disagree; 3 = disagree somewhat; 4 = agree somewhat; 5 = agree; 6 = strongly agree), which are divided into three specific self-efficacy subscales: (1) managing behaviours, (2) collaboration, and (3) using inclusive instruction. Each item is scored from one to six, with a higher score indicating greater self-efficacy to implement inclusive practices. Although the original instrument has high levels of validity and reliability, additional tests were performed to ensure its suitability for Cambodian samples, as shown in the 'Results' section. Subsequently, the general TEIP scale was incorporated into the analysis as the dependent variable in the two-level HLM.

4.4. Data collection and ethical issues

We obtained ethical approval and permission for data collection

from the Research Ethics Review Board of Hiroshima University in Japan and MoEYS in Cambodia. We completed the 'Plagiarism in the Humanities and Social Sciences' course (report #AP0000993110) on 17 October 2023, which was approved by the Association for the Promotion of Research Integrity. Following this, we submitted the research instrument to the Research Ethics Review Board for evaluation and received ethical approval (reference #HR-ES-000805) on 24 February 2023.

In Cambodia, we obtained permission from MoEYS to collect data. MoEYS officially approved the data collection request (reference #735) on 6 February 2023. We also obtained permission from the target schools to ensure that ethical concerns were addressed. We explained the study's purpose to the participants in advance and assured them that their personal information would be kept confidential. We explained that participation was voluntary, and they could withdraw at any time. We obtained individual consent from all participating primary school teachers and distributed the questionnaire to them.

4.5. Data Analysis

Multiple statistical methods were utilised for data analysis. Descriptive statistics, including the mean (M) and standard deviation (SD) with 99 % confidence interval (CI) estimation were calculated to assess the level of Cambodian TSE for inclusive practices. For the validity test, Principal Axis Factoring (PAF) with Equamax rotation was employed to perform a factor analysis on the 18-TEIP items to uncover its underlying structure. Furthermore, a reliability test (Cronbach's Alpha) was conducted to assess the internal consistency and reliability of the general self-efficacy scale and its subscales. We performed these analyses using the Statistical Package for Social Sciences (SPSS) software, version 23.

The primary analysis employed the two-level HLM to investigate the impact of individual- and school-level factors on teachers' general self-efficacy for inclusive practices. HLM is commonly used to analyse nested data with different hierarchical levels and accurately estimate the relationships between lower-level predictors and higher-level outcomes. It is employed to account for the shared variance in data with a hierarchical structure (Woltman et al., 2012).

In this study, the individual-level factors were nested in Level 1 of the hierarchies and encompassed 18 predictor variables, while the school-level factors were nested in Level 2 of the hierarchies and comprised two predictor variables, as presented in Table 5. The outcome variable was also included at Level 1. In principle, the outcome variable is nested in the lowest level of the hierarchies (Castro, 2002). Given that the outcome variable (general self-efficacy scale) was a continuous variable, a normal distribution was chosen for estimation within the two-level HLM.

Understanding the equations used at Level 1 and Level 2 is important for applying the two-level HLM. Below is a summary of the specified two-level HLM model with predictors used in the analysis.

Level-1 model:

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \beta_{2j}X_{2ij} + \dots + \beta_{Qj}X_{Qij} + r_{ij}$$

$$= \beta_{0j} + \sum_{q=1}^Q \beta_{qj}X_{qij} + r_{ij}$$

where

Y_{ij} is the outcome variable (self-efficacy) for teacher i nested within school j ;

β_{0j} is the Level-1 intercept;

β_{qj} ($q = 0, 1, \dots, Q$) are Level-1 regression coefficients;

X_{qij} is the Level-1 predictor q for teacher i in school j ;

r_{ij} is the Level-1 random effect. σ^2 is the variance of r_{ij} , that is the Level-1 variance. The random term is assumed as $r_{ij} \sim N(0, \sigma^2)$

(Raudenbush et al., 2019).

At Level 2, each of the Level-1 coefficients, β_{qj} , defined in the Level-1 model becomes an outcome variable in the Level-2 model, as shown below (Raudenbush et al., 2019; Woltman et al., 2012).

Level-2 model:

$$\beta_{qj} = \gamma_{q0} + \gamma_{q1}W_{1j} + \gamma_{q2}W_{2j} + \dots + \gamma_{qs_q}W_{s_qj} + u_{qj}$$

$$= \gamma_{q0} + \sum_{s=1}^{s_q} \gamma_{qs}W_{sj} + u_{qj}$$

where

γ_{q0} is the Level-2 intercept;

γ_{qs} ($q = 0, 1, \dots, S_q$) are Level-2 regression coefficients;

W_{sj} is a Level-2 predictor;

u_{qj} is the Level-2 random effect.

Mixed model:

$$Y_{ij} = \gamma_{00} + \sum_{s=1}^{s_q} \gamma_{0s}W_{sj} + u_{0j} + \sum_{q=1}^Q \left(\gamma_{q0} + \sum_{s=1}^{s_q} \gamma_{qs}W_{sj} + u_{qj} \right) X_{qij} + r_{ij}$$

Fixed effects:

γ_{00} is the overall intercept.

γ_{0s} are the coefficients for Level-2 predictors W_{sj} on the intercept.

γ_{q0} is the average effect of the q -th Level-1 predictor X_{qij} across all schools.

γ_{qs} are the coefficients for Level-2 predictors W_{sj} on the q -th Level-1 predictor X_{qij} .

Random effects:

u_{0j} is the random effect for the intercept in school j .

u_{qj} is the random effect for the q -th Level-1 predictor in school j .

r_{ij} is the residual error for teacher i in school j .

Before conducting the main analysis, a null model, known as the only-intercept model, was estimated to assess the suitability of the two-level HLM for this study (see Table 4). This estimation verifies that the variation in the outcome variable across Level-2 groups is significantly greater than zero and that HLM is necessary (Woltman et al., 2012). The intraclass correlation coefficient (ICC) was then computed to examine the variability in TSE attributed to the individual and school levels. In essence, the ICC is calculated to determine whether multilevel modelling is needed and reveals the deviance statistic along with other coefficients, serving as a baseline for subsequent, more intricate models (Garson, 2013).

The magnitude of ICC is calculated by dividing the intercept variance component of the model by the total variance components (Garson, 2013; Woltman et al., 2012). HLM evaluates the significance of the between-group variance (τ_{00}) but does not assess the significance of the within-group variance (σ^2) (Woltman et al., 2012). In the specified model, the total variance in Y_{ij} (self-efficacy) is partitioned into within-group and between-group components. Therefore, Variance (Y_{ij}) = Variance ($u_{qj} + r_{ij}$) = $\tau_{00} + \sigma^2$. The equation for calculating ICC is presented below. The results of the ICC of all models are presented in Table 5.

$$ICC = \frac{\tau_{00}}{\tau_{00} + \sigma^2}$$

where

σ^2 = Variance (r_{ij}) = within group variance in Y_{ij} (self-efficacy)

τ_{00} = Variance (u_{qj}) = between group variance in Y_{ij} (self-efficacy)

Furthermore, the final estimation of fixed effects, along with robust standard errors, was estimated and observed in model 1, model 2, and the mixed model, with standardised beta coefficients (β) and corresponding significance values (p). However, only the coefficients and significant values from the mixed model were used for interpretation and discussion in this study. We performed the analysis using HLM

software, version 8 with a standard licence.

5. Results

5.1. Validity and reliability of the instrument

The results of PAF models with Equamax rotation unveiled a total of three factors—(1) managing behaviours, (2) collaboration, and (3) using inclusive instruction—which were consistent with the study of Sharma et al. (2012). Kaiser-Meyer-Olkin (KMO) measure indicated the adequacy of sampling for the analysis ($KMO = 0.94$). Bartlett's test of sphericity was statistically significant ($\chi^2(136) = 9676.72, p < 0.001$), implying that the variables' correlations were sufficient for PAF. The communalities showed that a high percentage of the variance in the items was accounted for by the factors. Kaiser's criterion of eigenvalues greater than 1 confirmed a three-factor solution as the best fit, accounting for 56.77 % of the total variance.

Self-efficacy in managing behaviour (Factor 1), with six items, had an eigenvalue of 3.67 and accounted for 21.58 % of the variance. Self-efficacy in collaboration (Factor 2), with six items, had an eigenvalue of 3.29 and accounted for 19.36 % of the variance. Self-efficacy in using inclusive instruction (Factor 3), with five items, had an eigenvalue of 2.69 and accounted for 15.84 % of the variance. Overall, PAF provides support for the validity of the measure used in this study. Like the study of Sharma et al. (2012), three factors of TEIP were obtained, and had the same structure. It should be noted that item 6, 'I am confident in my ability to get students to work together in pairs or in small groups,' was excluded due to poor data fit under the PAF models.

Table 1 indicates that Alpha coefficients were 0.91 for self-efficacy in managing behaviours, 0.87 for self-efficacy in collaboration, 0.81 for self-efficacy in using inclusive instruction, and 0.93 for general self-efficacy scale. The Alpha coefficients of general self-efficacy scale and subscales were above 0.70, indicating sufficient internal consistency. The results of the validity and reliability tests suggest that the TEIP scale was reliable and suitable for measuring primary school TSE for inclusive practices in the context of Cambodia.

5.2. Teacher self-efficacy for inclusive practices

Table 2 shows that Cambodian teachers' general self-efficacy for inclusive practices was above the midpoint of the scale, which ranged from one to six ($M = 4.48, SD = 0.59$). It suggests that teachers reported high levels of self-efficacy to implement inclusive practices in Cambodian regular schools. The results show diverse and statistically significant scores of self-efficacy across the general self-efficacy scale and three subscales, as indicated by the non-overlapping 99 % CI values.

Teachers indicated the highest self-efficacy in managing students' behaviours in the regular schools ($M = 4.58, SD = 0.65$), which refers to the confidence in preventing and controlling students' disruptive behaviours to get them to follow classroom rules. The second optimal self-efficacy construct was the self-efficacy in using inclusive instruction for effective inclusion ($M = 4.46, SD = 0.72$). It refers to the ability to use assessment strategies, give simple explanations, design tasks that meet the individual needs of SWDs, and provide appropriate challenges for all students. However, teachers revealed the lowest self-efficacy in collaboration with their colleagues, parents, and other professionals such as special education teachers and therapists ($M = 4.39, SD = 0.69$).

Table 1
Results of the Reliability Test of General Self-Efficacy Scale and Its Subscales.

General self-efficacy scale and subscales	Cronbach's Alpha (α)
Managing behaviours	0.91
Collaboration	0.87
Using inclusive instruction	0.81
General self-efficacy scale	0.93

Table 2

Mean Scores (M) and Standard Deviations (SD) with 99 % CI Values of General Self-Efficacy Scale and Its Subscales (N = 1008).

General self-efficacy scale and subscales	M	SD	Lower CI	Upper CI	Sig. (p)
Managing behaviours	4.58	0.65	4.53	4.64	$p < 0.001$
Collaboration	4.39	0.69	4.34	4.45	$p < 0.001$
Using inclusive instruction	4.46	0.72	4.40	4.51	$p < 0.001$
General self-efficacy scale	4.48	0.59	4.43	4.53	$p < 0.001$

Table 3 shows the detailed distribution of all items within each self-efficacy subscale. In the construct of managing behaviours, the item 'I am able to get children to follow classroom rules' (Item 10) had the highest mean score ($M = 4.78, SD = 0.69$), while the item 'I am able to calm a student who is disruptive or noisy' (Item 9) had the lowest mean score ($M = 4.51, SD = 0.81$).

In the construct of using inclusive instruction, the item 'I am able to provide an alternate explanation or example when students are confused' (Item 2) had the highest mean score ($M = 4.68, SD = 0.89$), whereas the item 'I am confident in designing learning tasks so that the individual needs of SWDs are accommodated' (Item 3) had the lowest mean score ($M = 4.18, SD = 1.06$).

In the construct of collaboration, the item 'I can make parents feel comfortable coming to school' (Item 16) had the highest mean score ($M = 4.66, SD = 0.75$), while the item 'I am confident in my ability to get parents involved in school activities of their children with disabilities' (Item 15) had the lowest mean score ($M = 4.24, SD = 0.93$).

Table 3

Mean Scores (M) and Standard Deviations (SD) of Included TEIP Items (see Sharma et al., 2012).

Self-efficacy subscales and items	M	SD
<i>Self-efficacy in using inclusive instruction</i>		
1. I can use a variety of assessment strategies (for example, portfolio assessment, modified tests, performance-based assessment, etc.).	4.23	1.07
2. I am able to provide an alternate explanation or example when students are confused.	4.68	0.89
3. I am confident in designing learning tasks so that the individual needs of SWDs are accommodated.	4.18	1.06
4. I can accurately gauge student comprehension of what I have taught.	4.59	0.85
5. I can provide appropriate challenges for very capable students.	4.60	0.88
<i>Self-efficacy in managing behaviours</i>		
7. I am confident in my ability to prevent disruptive behaviour in the classroom before it occurs.	4.56	0.81
8. I can control disruptive behaviour in the classroom.	4.60	0.81
9. I am able to calm a student who is disruptive or noisy.	4.51	0.81
10. I am able to get children to follow classroom rules.	4.78	0.69
11. I am confident when dealing with students who are physically aggressive.	4.54	0.81
12. I can make my expectations clear about student behaviour.	4.52	0.76
<i>Self-efficacy in collaboration</i>		
13. I can assist families in helping their children do well in school.	4.36	0.88
14. I am able to work jointly with other professionals and staff (e.g., aides, other teachers) to teach SWDs in the classroom.	4.48	0.86
15. I am confident in my ability to get parents involved in school activities of their children with disabilities.	4.24	0.93
16. I can make parents feel comfortable coming to school.	4.66	0.75
17. I can collaborate with other professionals (e.g., itinerant teachers or speech pathologists) in designing educational plans for SWDs.	4.34	0.95
18. I am confident in informing others who know little about laws and policies relating to the inclusion of SWDs	4.29	0.94

Note. Item 6 was excluded due to poor data fit under the PAF models.

5.3. Effects of individual- and school-level factors

Table 4 presents the results of the null model of two-level HLM, indicating the suitability of the model and variability in TSE for inclusive practices. It was found that there were significant variations in TSE for inclusive practices across observed schools ($\chi^2(235) = 428.93, p < 0.001$). The results of the ICC of the null model indicated that 84 % of the variance in TSE was attributable to individual-level factors at Level 1, while 16 % was attributable to school-level factors at Level 2. It was observed that the large magnitude of the variance was collectively observed across schools, thus justifying the use of the two-level HLM for the data of the current study.

Table 5 shows that at individual level, the results of the final estimation of fixed effects with robust standard errors of the mixed model revealed that four predictor variables such as gender, interaction experience, experience of IET, and knowledge of disabilities significantly predicted TSE for inclusive practices. It was found that gender was a significant predictor and had a negative effect on TSE, indicating that female teachers perceived lower self-efficacy for IE practices than their male counterparts ($\beta = -0.11, p < 0.01$). The mixed model shows that for each one-standardised unit increase in the proportion of females, there was a significant decrease of 0.11 standardised units in general self-efficacy.

Moreover, interaction experience with PWDs was a significant predictor and positively predicted TSE ($\beta = 0.12, p < 0.05$), indicating that for each one-standardised unit increase in experience interacting with PWDs, there was a significant increase of 0.12 standardised units in general self-efficacy. The results further reveal that experience of IET significantly and positively predicted TSE for inclusive practices ($\beta = 0.10, p < 0.01$). It indicates that for each one-standardised unit increase in experience of IET, there was a significant increase of 0.10 standardised units in general self-efficacy. Lastly, knowledge of disabilities was also a significant predictor for general self-efficacy ($\beta = 0.09, p < 0.05$). For each one-standardised unit increase in knowledge of disabilities, there was a significant increase of 0.09 standardised units in general self-efficacy to implement IE.

At school level, the results of the mixed model show that implementation of IE programmes significantly and positively predicted TSE. It was found that teachers from schools that implemented IE programmes had significantly greater general self-efficacy in implementing inclusive practices than those from schools that had not ($\beta = 0.11, p < 0.05$). For each one-standardised unit increase in the proportion of IE programmes, there was a significant increase of 0.11 standardised units in general self-efficacy to implement inclusive practices, as shown in Table 5 below. These findings are discussed with a call for implications to improve Cambodian TSE in teaching SWDs in the context of Cambodia.

6. Discussion

6.1. Teacher self-efficacy for inclusive practices

6.1.1. Managing students' behaviours

This study found that Cambodian teachers generally expressed strong self-efficacy for inclusive practices. However, in the construct of managing behaviours, teachers expressed the lowest self-efficacy in

Table 4
Variability in Self-Efficacy between Individual and School Levels.

Units of analysis	Variance	Variance explained (%)
Between individuals ($N = 1008$)	0.29	84 ^a
Between schools ($N = 236$)	0.06	16 ^b

Note. ^aProportion of the variance explained (between individuals) = $0.29/(0.06 + 0.29) = 0.84$; ^bProportion of the variance explained (between schools) = $0.06/(0.06 + 0.29) = 0.16$

Table 5

Results of the Two-Level Hierarchical Linear Model Assessing Factors Influencing Teachers' Self-Efficacy for Inclusive Practices ($N = 1008$).

Independent variables	Null model coefficient	Level-1 coefficient	Level-2 coefficient	Mixed-model coefficient
Final intercept	4.48***	4.48***	4.48***	4.48***
<i>Individual-level factors (Level-1 hierarchy)</i>				
Gender		-0.11**		-0.11**
Age		0.01		0.01
Disability		0.01		0.01
Interaction with PWDs		0.12*		0.12*
Family or relatives with disabilities		-0.04		-0.04
Teacher training programmes		-0.02		-0.02
Experience of IET		0.10**		0.10**
Adequate IET		0.05		0.05
Knowledge of PECD		0.01		0.01
Knowledge of PIE		-0.09		-0.09
Knowledge of IE		0.08		0.08
Knowledge of special education		0.02		0.02
Knowledge of disabilities		0.09*		0.09*
Knowledge of IEPs		0.04		0.04
Years of teaching		0.01		0.01
Experience of teaching SWDs		0.05		0.05
Confidence in teaching SWDs		0.04		0.04
Collaboration experience		0.07		0.07
<i>School-level factors (Level-2 hierarchy)</i>				
Schools implementing IE			0.11*	0.11*
School location			0.10	0.10
Between-individual variance	84 %	82 %	85 %	82 %
Between-school variance	16 %	18 %	15 %	18 %
Total variance explained	100 %	100 %	100 %	100 %

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

dealing with disruptive or noisy students. This finding is consistent with previous studies by Yada and Alnahdi (2021), Alnahdi (2019), and Monteiro et al. (2019).

In their comparative study, Yada and Alnahdi (2021) found that Saudi Arabian participants expressed significantly lower general self-efficacy for inclusive practices compared to Japanese participants, specifically regarding their ability to calm disruptive or noisy students in inclusive classrooms. Similarly, Alnahdi (2019) and Monteiro et al. (2019) found that although their participants displayed high levels of general self-efficacy for inclusive practices, they expressed less confidence in dealing with students exhibiting aggressive behaviours and preventing disruptive behaviour in the classroom.

In Cambodia, the management of students' disruptive and noisy behaviours presents significant challenges in regular schools, primarily due to large class sizes. According to statistics from the EMIS, the student-teacher ratio at primary level was the highest among all educational levels, averaging 45.1, with urban areas at 27.9 and rural areas at 36.7 in the academic year 2021–2022 (MoEYS, 2022b). This situation results in detrimental effects on all students, including those with and without disabilities, as it reduces their ability to concentrate on the lesson and decreases their involvement in the teaching and learning process. It is difficult for teachers to keep them on track (Horne &

Timmons, 2009; Round et al., 2016).

To address these challenges, it is essential to maintain smaller class sizes in inclusive classrooms. Smaller classes allow teachers to give more individualised attention to students, especially those with disabilities, and reduce the time spent dealing with disruptive and noisy behaviour (Organisation for Economic Co-operation and Development (OECD), 2016). Moreover, Monsen et al. (2014) argue that promoting an adaptive classroom environment helps teachers address these concerning behaviours and improve students' learning outcomes in inclusive classroom settings. This highlights the ability of teachers to create classroom environments that enhance student satisfaction and unity while minimising factors such as friction, competition, and other challenges (Monsen et al., 2014).

6.1.2. Using inclusive instruction

In the construct of using inclusive instruction, Cambodian teachers demonstrated the highest self-efficacy in providing alternative explanations or examples when students are confused. However, they expressed the lowest self-efficacy in designing learning tasks that catered to the individual needs of SWDs. This finding aligns with the results of previous studies by Yada and Savolainen (2017) and Alnahdi (2019).

Alnahdi (2019) found that Saudi Arabian teachers had the highest confidence in providing alternate explanations but the least confidence in designing learning tasks for SWDs. Similarly, Yada and Savolainen (2017) found that their Japanese participants expressed more confidence in providing alternate explanations than in designing learning tasks for SWDs. This consistent finding strongly highlights the negative consequences of inadequate knowledge of disabilities and a lack of training on teachers' ability to effectively implement inclusive instruction in inclusive classrooms (Alnahdi, 2019).

The significant number of SWDs in regular primary schools in Cambodia increases the likelihood that Cambodian teachers encounter SWDs in their classrooms. While regular schools in Cambodia generally accept various types of disabilities, with mild condition (Kalyanpur, 2016; Nishio, 2019), teachers' limited knowledge of disabilities and inadequate skills in designing learning tasks to meet individual needs can have detrimental impacts on SWDs. In the current study, most teachers rated their knowledge of disabilities as 'average' (51.29 %) and 'poor' (42.26 %). This lack of expertise leads to a consequence that aligns with the findings of previous studies, indicating that Cambodian SWDs have been marginalised within regular educational settings and experienced a high dropout rate before completing primary education (Nishio, 2019; MoP, 2020; Ravet & Mtika, 2021).

To address these challenges, teacher preparation and continuous professional development programmes should focus on enhancing knowledge of disabilities and developing skills in designing learning tasks for diverse disabilities. Teachers also need to acquire knowledge in designing IEPs, which enables them to meet the individual needs of SWDs (Musyoka & Diane Clark, 2017). In this study, most teachers rated their knowledge of IEPs ($M = 2.37$, $SD = 0.65$) as one of the lowest among the knowledge related to IE. This suggests that training programmes should place a strong emphasis on the ability to design and implement IEPs. The active involvement of parents in the development and implementation process of IEPs is imperative to support effective inclusion (Musyoka & Diane Clark, 2017). Moreover, both pre-service and in-service training programmes should integrate hands-on experiences that allow teachers to work directly with SWDs during their teaching practice (Alnahdi, 2019).

6.1.3. Collaboration

In this study, Cambodian teachers rated collaboration as their area of lowest self-efficacy among the three self-efficacy subscales. Specifically, they reported the lowest self-efficacy in getting parents involved in school activities for their children with disabilities, which also aligns with the findings of Alnahdi (2019) and Yada and Savolainen (2017).

These studies suggest that teachers may not involve parents in school activities to support SWDs due to concerns about their own capabilities, limited time, heavy workloads, and the additional efforts required. Teachers have multiple responsibilities beyond teaching, which often require them to dedicate more time to extracurricular activities and leave less time for collaborating with parents (Yada & Savolainen, 2017).

In Cambodia, these challenges extend to both teachers and parents. Firstly, there are challenges related to parental collaboration. As a predominantly Buddhist country, Cambodian has traditional beliefs that disabilities are caused by sins committed in past lives. This belief brings shame to families and leads them to conceal their children's disabilities, creating significant barriers for children with disabilities to access educational services (Disability Action Council, 2003; Kalyanpur, 2014).

Parents of children with disabilities often hesitate to enrol their children in regular schools, believing that regular schools are ill-equipped to meet their children's learning needs, particularly those in rural areas of Cambodia (Ravet & Mtika, 2021; Woodcock et al., 2022). Their lack of engagement with schools was attributed to their inability to provide support to their children with disabilities at home or in school due to their limited literacy and numeracy skills, as reported in the study by Ravet & Mtika (2021).

To improve teacher-parent collaboration, teachers working in inclusive classrooms need more opportunities to engage with parents of children with disabilities compared to their counterparts working in regular classrooms (Horne & Timmons, 2009; Malinen et al., 2013). School principals can support this process by allocating adequate time for teachers to collaborate with parents in planning additional academic support activities for SWDs (Horne & Timmons, 2009).

This study further found that Cambodian teachers had limited knowledge and experience in IE and lacked adequate training to work effectively in inclusive classroom settings. This results in a lack of understanding and confidence in their roles and responsibilities within the collaborative support process necessary for developing inclusive classrooms (Malinen et al., 2013), leading them to hold a strong belief that SWDs should be educated in segregated education settings (Woodcock et al., 2022). Pijl (2010) argues that facilitating teachers to glean insights from their individual experiences and foster collaboration among colleagues at the school level yields a transient yet substantial enhancement in TSE and the assurance to apply inclusive practices (Ekins et al., 2016; Pijl, 2010). This can be achieved through regular meetings with colleagues to deliberate on their experiences, decisions, the underlying rationales, and their consequences (Pijl, 2010).

6.2. Effects of Individual- and School-Level Factors

6.2.1. Gender

The results of the two-level HLM showed that gender was a significant predictor. Particularly, male teachers demonstrated higher self-efficacy for teaching SWDs compared to their female counterparts, despite the fact that only 33 % of the participants were male and 67 % were female.

Van Mieghem et al. (2022) found that in Belgium, female teachers had higher general self-efficacy in implementing inclusive practices with students without SEND, but lower self-efficacy for teaching students with SEND. On the other hand, male teachers had significantly higher general self-efficacy for teaching students with SEND. In Hong Kong, the results of pre-course and post-course assessments of IE showed that male teachers expressed greater self-efficacy for inclusive practices across all constructs compared to their female counterparts (Forlin & Sin, 2010).

Research has asserted that female teachers perceive workloads and challenges associated with managing students' disruptive behaviour as occupational stress, leading to significantly higher concerns about their self-efficacy and ultimately reducing job satisfaction compared to male teachers (Klassen & Chiu, 2010; Tsakiridou & Polyzopoulou, 2014). Moreover, given the social norms that associate men with having more

firm control in managing behaviours, male teachers tended to have stronger confidence in managing students' behaviours compared to their female counterparts in inclusive classrooms (Specht & Metsala, 2018).

While previous studies have shown that female teachers tend to perceive lower self-efficacy, most have not provided specific recommendations to enhance their self-efficacy. This study suggests that a collaborative model at the school and classroom levels can be instrumental in boosting the self-efficacy of female teachers. This model relies on three types of cooperative resources: physical support, human support, and the school environment (Avramidis & Norwich, 2002). Van Mieghem et al. (2022) contend that increased utilisation of cooperative resources enhances teachers' efficacy in implementing inclusive practices. Notably, human support sources, including co-teaching or team teaching, have a significant impact on increasing TSE. Such efficacy can only be developed through close collaboration between teachers or between a teacher and an internal or external counsellor. In this context, it is vital for female teachers to receive focused support from their male colleagues and counsellors (Mieghem et al., 2022).

Leadership teams also play a crucial role in this process. They should actively encourage and prioritise the participation of female teachers in such collaborative and counselling efforts whenever opportunities arise. Leadership teams should prioritise pedagogical inclusivity, embrace diversity, and promote teacher self-reflection to cultivate an inclusive culture within schools (Valdés & Gómez-Hurtado, 2023).

6.2.2. Interaction experience

This study revealed that teachers with previous experience interacting with PWDs had higher self-efficacy in implementing inclusive practices compared to those without such experience. Most of them (69 %) reported having experience interacting with PWDs.

Wray et al. (2022) identified interaction experience as a powerful predictor influencing TSE for inclusive practices in 41 out of 71 selected studies. Different forms of interaction experience, such as through friends, family, community, and workplaces, have been found to significantly enhance TSE (Crowson & Brandes, 2014; Malinen et al., 2013; Specht & Metsala, 2018; Wray et al., 2022; Yada & Savolainen, 2019).

In the Cambodian context, teachers are likely to encounter and interact with PWDs, as the country has been identified as one of the countries with the highest rates of PWDs worldwide (UNICEF, 2003). Some studies found that these experiences indirectly impacted TSE for inclusive practices (e.g., Crowson & Brandes, 2014; Yada & Savolainen, 2019). However, concerning Cambodian teachers, the results of the two-level HLM supported most previous studies (e.g., Crowson & Brandes, 2014; Malinen et al., 2013; Specht & Metsala, 2018; Yada & Savolainen, 2019), revealing that interaction experience had a direct effect on their self-efficacy.

Interaction of teachers with individuals with disabilities serves as a potential source of mastery experience, even when these interactions occur outside the school context (Malinen et al., 2013). Changes in policies and legislation aimed at ensuring and promoting the rights of individuals with disabilities alter the living and working environments to accommodate vicarious experience. This creates an enabling atmosphere for more active interaction between people with and without disabilities in various aspects of the society (Wray et al., 2022).

In Cambodia, the participation of PWDs in the economic, social, political, and cultural spheres remains a critical challenge (UNESCO, 2005). As Cambodia actively works towards transforming into a fully IE system, policies and legislation should aim at promoting IE concepts and practices at all educational levels. Wray et al. (2022, p.11) claim that 'where our societies work to be more inclusive in general and at all stages of life, we increase the likelihood of building TSE for IE'.

6.2.3. Inclusive education training

This study revealed that the experience of IET was the most powerful predictor of TSE. It had a highly positive impact on Cambodian TSE for

inclusive practices, despite most participants reporting limited exposure to such training. They expressed dissatisfaction with the inadequacy and ineffectiveness of the IET they had received (72 %) and maintained a strong desire to participate in such training as part of their professional learning (89 %). A large body of existing literature supports these findings, indicating that IET significantly influences the self-efficacy of both pre-service and in-service teachers to implement inclusive practices across various contexts (e.g., Loreman et al., 2013; Malinen et al., 2013; Sharma & Sokal, 2015).

Loreman et al. (2013) conducted a study across four countries and found that experience of IET had a significant impact on TSE, especially when combined with high levels of knowledge of laws and policies related to IE. Similarly, a study examining pre-service teachers' attitudes, concerns, and self-efficacy towards inclusion in Australia and Canada reported that teachers from both countries showed significant improvements in their self-efficacy following the completion of an IET course (Sharma & Sokal, 2015).

For in-service teacher education, Malinen et al. (2013) found that the amount of IET significantly influenced Finnish TSE to implement inclusive practices. However, the same predictor did not yield significant results for Chinese and South African models. Malinen et al. (2013) claimed that this was not a surprised finding. This disparity was attributed to Finland's longer history of inclusive practice implementation and better resourced for inclusion with well-developed learning support, as opposed to China and South Africa.

In Cambodia, several studies have commonly noted that the country has a relatively recent history of implementing IE. This has resulted in a rudimentary understanding of both the concept and the practical application of IE among practitioners (Kalyanpur, 2011, 2014, 2016; Nishio, 2019). Schools are not accessible for SWDs and lack adequate resources to support inclusion (Ravet & Mtika, 2021; Šiška & Suchánek, 2015). In-service IET fail to equip teachers with sufficient and practical knowledge and skills to include SWDs (Kuroda et al., 2017; Ravet & Mtika, 2021; Šiška & Suchánek, 2015). The training has insufficient on-site support for effective inclusion (King, 2018; Kuroda et al., 2017).

Insufficient duration of training, inadequate expertise of trainers, and ineffective facilitation of the training course also contribute to poor quality of IET. These factors lead to teachers' lack of readiness, resulting in insufficient knowledge, skills, and confidence to effectively work in inclusive classrooms (Šiška & Suchánek, 2015). The prevalent cascade training model also poses a critical concern about the explicit content delivery and sustainability of the training at different hierarchical levels, as it tends to weaken as it cascades down from national level or provincial level to school level. This has been asserted as 'a watering down the further down you go' (King, 2018, p. 9).

For pre-service teacher education, the curriculum of the new teacher training programme (12+4) and the teacher training guide for IE indicate a notable deficiency in IET. Like the previous 12+2 programme, the new curriculum of 12+4 programme does not prioritise IE and considers it as an 'optional extra' topic within a broader 'General Topic in Education' course. The course accounts for two credits or 30 hours of instruction time and is provided in the first year of training. However, the topic on basic concept of IE is allocated approximately only six hours, even though the teacher training guide outlines 28 hours and 25 minutes of instruction time for completing the IE course (see MoEYS, 2017a, 2017b). These issues likely raise teachers' negative attitudes towards inclusion and lack of confidence in designing learning tasks to meet the needs of SWDs, resulting in low self-efficacy for inclusive practices (Kuroda et al., 2017).

These findings highlight the necessity of effective IET within pre-service training programmes at all educational levels. It is strongly suggested that an IE course be made a mandatory module in pre-service teacher education curricula (Ekins et al., 2016). Moreover, in-service trainings should be provided to support teachers with little or no experience of IET, as well as those working in schools that implement IE programmes (Kuroda et al., 2017; Ravet & Mtika, 2021). The trainings

should integrate on-site support, involving assistance from experts such as special education teachers, to foster teachers' confidence and self-efficacy in implementing effective inclusive practices (King, 2018; Kuroda et al., 2017). Expanding IE programmes can also provide regular teachers with valuable practical experience in teaching SWDs (Alnahdi, 2019). Even with limited or without experience of IET, opportunities to teach SWDs allows teachers to gain hands-on experience, thereby increasing their confidence to effectively implement inclusive practices in their classrooms (Shah et al., 2016).

6.2.4. Knowledge of disabilities

In this study, among all predictors related to knowledge, only knowledge of disabilities significantly and positively predicted TSE. Most participants rated their knowledge of disabilities as 'average' (51 %) and 'poor' (42 %). This finding is consistent with Ekins et al. (2016) and Sharma and Sokal (2015).

Similarly, Sharma and Sokal (2015) found that about 63 % of their participants reported their knowledge of disabilities as 'nil' or 'poor', and 37 % indicated it as 'average'. Ekins et al. (2016) investigated the effects of teachers' knowledge pertaining to SEND on their self-efficacy and found that knowledge of disabilities was one of the strongest predictors. Knowledge of disabilities is a common concerning area, which worries teachers about their ability to implement inclusion and significantly influences their self-efficacy to implement inclusive practices (Ekins et al., 2016; Sharma & Sokal, 2015).

In the Cambodian context, this issue partly results from the implementation gaps in policies and practices concerning the education of SWDs at both national and sub-national levels. At the national level, the implementation of policies lacks a robust and systematic approach (MoEYS, 2022a). A policy monitoring report in 2020 by MoEYS (2020) revealed that despite the existence of PECD for over a decade, only 5 % of schools reported implementing it. Furthermore, some educational staff (16.55 %) at the school level expressed a lack of capacity to implement inclusive practices in regular schools due to resource constraints and a shortage of qualified teachers to teach SWDs, while others (35.25 %) were not even aware of the existence of PECD.

The absence of on-site support at the school level further inhibits teachers' confidence in implementing changes to support SWDs (King, 2018). To address the discrepancy between policies and practices, a systematic dissemination system and an effective training model should be established. Additionally, teachers need to have a comprehensive understanding of relevant laws, policies, and available support mechanisms for IE (Alnahdi, 2019; Sharma et al., 2008).

As previously discussed, the training guide has been developed as a key resource for both pre-service and in-service training programmes. While this guide encompasses lessons grounded in various theories such as universal design for learning, a holistic approach, and a twin-track approach, it places less emphasis on the practical aspects of IE. Specifically, it does not adequately cover the implementation of accommodation and modification strategies, the acquisition of skills to identify individual learning needs, and the essential knowledge about disabilities (MoEYS, 2017c). Integrating of knowledge and understanding of disabilities should now be an indispensable component of teacher education and continuous professional development programmes. This integration should be viewed as a fundamental element of teachers' roles and skill development, rather than as an 'optional addition' to the training curricula (Ekins et al., 2016).

6.2.5. Inclusive education programmes

This study found a significant impact of IE programmes on TSE for inclusive practices at Cambodian primary schools. Teachers in primary schools that implemented IE programmes reported higher self-efficacy in teaching SWDs than those who did not. This finding is consistent with a substantial body of previous research (e.g., Malinen et al., 2013; Monteiro et al., 2019; Yada & Savolainen, 2019).

Effective IE programmes provide teachers with practical experience

in teaching SWDs. As noted earlier, such experience directly impacts TSE and their attitudes towards IE (Wray et al., 2022). In this study, all primary schools investigated had a history of including SWDs, and 50 % implemented IE programmes at the time of data collection. However, only 45 % of teachers reported having some experience in teaching SWDs. This disparity may stem from the belief that SWDs should be taught by more experienced teachers who are better prepared for this role. Providing such hands-on experience to all teachers enhances their self-efficacy and attitudes towards implementing inclusive practices, especially for novice teachers. This, in turn, positively influences the development of IE system at the school level (Alnahdi, 2019; Wray et al., 2022).

Research shows that the exposure to inclusive classroom settings and extended opportunities to teach SWDs strongly foster TSE (Peebles & Mendaglio, 2014; Specht & Metsala, 2018). In schools with IE programmes, it is essential for every teacher to have opportunities to teach SWDs. This can be achieved by rotating teaching assignments between regular and inclusive classrooms across different grade levels. Implementing such an approach requires strong collaborative support from experienced teachers and active involvement from all school staff (Pijl, 2010), especially the school leadership teams (Horne & Timmons, 2009; Valdés & Gómez-Hurtado, 2023). However, given the limited availability of IE programmes and IET opportunities in Cambodia (Kuroda et al., 2017; Ravet & Mtika, 2021), a school-based mentoring and coaching programme serves as an alternative and effective method to enhance teachers' confidence and abilities when working collaboratively within inclusive classroom settings (Pijl, 2010; Wray et al., 2022).

7. Limitations and future studies

There are some limitations to this study. Data were collected through self-reported questionnaires, which may reflect the subjective views of teachers. Consequently, the findings should be approached with caution. Additionally, the study focused on primary school teachers, limiting the generalisability of the results to only primary education. Research indicates that different educational settings, such as primary and secondary schools, can significantly impact TSE (e.g., Specht et al., 2016). Therefore, future studies should consider collecting data from teachers at other educational levels. Additionally, there were a limited number of school-level predictors in this study. Future research may benefit from employing multiple instruments beyond a single questionnaire to capture a broader range of school-related predictors.

8. Conclusion

This study revealed notable variations among the three subscales, with Cambodian primary school teachers reporting the highest self-efficacy in managing students' behaviour and the lowest self-efficacy in collaboration. Further examination of the individual items within these subscales demonstrated that teachers exhibited the least self-efficacy in addressing disruptive or noisy student behaviours, designing learning tasks tailored to the unique needs of SWDs, and involving parents in school activities. Importantly, the analysis identified several factors that significantly predicted their general self-efficacy, including gender, experience of interacting with PWDs, experience in IET, knowledge of disabilities, and the presence of IE programmes at the school level. This research underscores the multifaceted implications essential for the successful implementation of IE in Cambodia.

The findings highlight the critical role of classroom size, adaptive classroom environments, knowledge of disabilities, proficiency in designing diverse learning tasks, comprehension of IEPs, ability to actively involve parents, hands-on experience, allocation of sufficient time for teacher-parent collaboration, and fostering teachers' confidence in their roles and responsibilities within the collaborative support process. Moreover, it has become evident that the enhancement of

human sources of support and an increase in participation, especially among females, are vital for promoting an inclusive culture within schools.

The evolution of policies and legislation aimed at fostering the participation of PWDs plays a pivotal role. A shift towards an improved cascade training model, with heightened emphasis on on-site support, proves to be instrumental. Lastly, making IE courses mandatory in teacher education at all levels, accompanied by the development of school-based mentoring and coaching programmes, paves the way for a more inclusive educational landscape in Cambodia, thus ensuring equal access and quality education for all students, especially those with disabilities.

Ethics statement

This study was approved by the Research Ethics Review Board of Hiroshima University with reference number 'HR-ES-000 805' on 24 February 2023.

Appendix 1

Mean Scores (M), Standard Deviations (SD), Number (n), and Percentages (%) of Variables

Variables	n	%	M	SD
<i>Dependent variable</i>				
General TSE scale			4.48	0.59
<i>Independent variables</i>				
Gender				
Male	336	33		
Female	672	67		
Age				
18–24 years	162	16		
25–31 years	281	28		
32–38 years	222	22		
39–45 years	133	13		
46–52 years	121	12		
53–59 years or more	89	9		
Disability				
No	964	96		
Yes	44	4		
Interaction with PWDs				
No	321	32		
Yes	687	68		
Family or relatives with disabilities				
No	832	83		
Yes	176	18		
Teacher training programmes				
lower than grade 9	202	20		
9+2 programme	188	19		
12+2 programme	595	59		
12+4 programme	23	2		
Experience of IET				
No	704	70		
Yes	304	30		
Adequate IET			2.24	0.75
Knowledge of PECD			2.53	0.60
Knowledge of PIE			2.54	0.59
Knowledge of IE			2.57	0.60
Knowledge of special education			2.32	0.64
Knowledge of disabilities			2.51	0.62
Knowledge of IEPs			2.37	0.65
Years of teaching				
1–5 years	275	27		
6–10 years	218	22		
11–15 years	149	15		
16–20 years	92	9		
21–25 years	77	8		
26–30 years	73	7		
31–35 years	83	8		
36–40 years or more	41	4		

(continued on next page)

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Norimune Kawai: Writing – review & editing, Supervision. **Sokunrith Pov:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rie Murakami:** Writing – review & editing, Supervision.

Conflict of Interest

We declare no competing interests.

(continued)

Variables	n	%	M	SD
Experience of teaching SWDs				
No	558	55		
Yes	450	45		
Confidence in teaching SWDs			2.76	0.66
Collaboration experience				
No	473	47		
Yes	535	53		
Schools implementing IE programmes				
No	503	50		
Yes	505	50		
School location				
Urban	296	29		
Rural	712	71		

Appendix 2

Correlation Coefficients

Independent variables	General self-efficacy scale
Gender	−0.12***
Age	0.02
Disability	0.06
Interaction with PWDs	0.18***
Family or relatives with disabilities	0.03
Teacher training programmes	−0.07*
Experience of IET	0.09**
Adequate IET	0.13***
Knowledge of PECD	0.15***
Knowledge of PIE	0.18***
Knowledge of IE	0.21***
Knowledge of special education	0.16***
Knowledge of disabilities	0.23***
Knowledge of IEPs	0.18***
Years of teaching	0.02
Experience of teaching SWDs	0.16***
Confidence in teaching SWDs	0.25***
Collaboration experience	0.21***
Schools implementing IE programmes	0.16***
School location	0.10**

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Data Availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

- Alnahdi, G. (2019). Are we ready for inclusion? Teachers' perceived self-efficacy for inclusive education in Saudi Arabia. *International Journal of Disability, Development and Education*, 67(2), 182–193. <https://doi.org/10.1080/1034912X.2019.1634795>
- Avramidis, E., & Norwich, B. (2002). Teachers' attitudes towards integration/inclusion: a review of the literature. *European Journal of Special Needs Education*, 17(2), 129–147. <https://doi.org/10.1080/08856250210129056>
- Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioural change. *Psychological Review*, 84(2), 191. <https://doi.org/10.1037/0033-295X.84.2.191>
- Cambodia National Assembly. (2007). *The law on education*. Cambodia: Phnom Penh.
- Castro, S. L. (2002). Data analytic methods for the analysis of multilevel questions: A comparison of intraclass correlation coefficients, rwg (j), hierarchical linear modelling, within-and between-analysis, and random group resampling. *The Leadership Quarterly*, 13(1), 69–93. [https://doi.org/10.1016/S1048-9843\(01\)00105-9](https://doi.org/10.1016/S1048-9843(01)00105-9)
- Chao, C. N. G., Forlin, C., & Ho, F. C. (2016). Improving teaching self-efficacy for teachers in inclusive classrooms in Hong Kong. *International Journal of Inclusive Education*, 20(11), 1142–1154. <https://doi.org/10.1080/13603116.2016.1155663>
- Crowson, H. M., & Brandes, J. A. (2014). Predicting pre-service teachers' opposition to inclusion of students with disabilities: A path analytic study. *Social Psychology of Education*, 17, 161–178. <https://doi.org/10.1007/s11218-013-9238-2>
- Disability Action Council. (2003). *Inclusive education training in Cambodia*. Cambodia: Phnom Penh.
- Ekins, A., Savolainen, H., & Engelbrecht, P. (2016). An analysis of English teachers' self-efficacy in relation to SEN and disability and its implications in a changing SEN policy context. *European Journal of Special Needs Education*, 31(2), 236–249. <https://doi.org/10.1080/08856257.2016.1141510>
- Emmers, E., Baeyens, D., & Petry, K. (2020). Attitudes and self-efficacy of teachers toward inclusion in higher education. *European Journal of Special Needs Education*, 35(2), 139–153. <https://doi.org/10.1080/08856257.2019.1628337>
- Forlin, C., & Lian, M. G. J. (2008). *Reform, inclusion and teacher education: Towards a new era of special education in the Asia-Pacific region (1st ed.)*. Routledge.
- Forlin, C., & Sin, K. F. (2010). Developing Support for inclusion: a professional learning approach for teachers in Hong Kong. *International Journal of Whole Schooling*, 6(1), 7–26.
- Fowler, F. J. (2009). *Survey research methods*. SAGE Publications, Inc.. <https://doi.org/10.4135/9781452230184>
- Garson, G.D. (2013). *Introductory guide to HLM with HLM 7 software*. Sage Los Angeles, CA.
- Horne, P. E., & Timmons, V. (2009). Making it work: Teachers' perspectives on inclusion. *International Journal of Inclusive Education*, 13(3), 273–286. <https://doi.org/10.1080/13603110701433964>
- Kalyanpur, M. (2011). Paradigm and paradox: Education for all and the inclusion of children with disabilities in Cambodia. *International Journal of Inclusive Education*, 15(10), 1053–1071. <https://doi.org/10.1080/13603116.2011.555069>
- Kalyanpur, M. (2014). Distortions and dichotomies in inclusive education for children with disabilities in Cambodia in the context of globalisation and international development. *International Journal of Disability, Development and Education*, 61(1), 80–94. <https://doi.org/10.1080/1034912X.2014.878546>
- Kalyanpur, M. (2016). Inclusive education policies and practices in the context of international development. Lessons from Cambodia. *ZEP: Zeitschrift für Internationale Bildungsforschung und Entwicklungspädagogik*, 39(3), 16–21. <https://doi.org/10.25656/01:15452>
- King, E.F. (2018). Developing teacher capacity in Cambodia: An expanded model. *Asian Education and Development Studies*. <https://doi.org/10.1108/AEDS-06-2017-0053>

- Klassen, R. M., & Chiu, M. M. (2010). Effects on teachers' self-efficacy and job satisfaction: Teacher gender, years of experience, and job stress. *Journal of Educational Psychology*, 102(3), 741. <https://doi.org/10.1037/a0019237>
- Kuroda, K., Kartika, D., & Kitamura, Y. (2017). *Implications for teacher training and support for inclusive education in Cambodia: An empirical case study in a developing country*. JICA Research Institute.
- Kuyini, A. B., Desai, I. (Ishwar), & Sharma, U. (2020). Teachers' self-efficacy beliefs, attitudes and concerns about implementing inclusive education in Ghana. *International Journal of Inclusive Education*, 24(14), 1509–1526. <https://doi.org/10.1080/13603116.2018.1544298>
- Loireman, T., Sharma, U., & Forlin, C. (2013). Do pre-service teachers feel ready to teach in inclusive classrooms? A four country study of teaching self-efficacy. *Australian Journal of Teacher Education (Online)*, 38(1), 27–44. <https://search.informit.org/doi/10.3316/informit>
- Malinen, O. P., Savolainen, H., Engelbrecht, P., Xu, J., Nel, M., Nel, N., & Tlale, D. (2013). Exploring teacher self-efficacy for inclusive practices in three diverse countries. *Teaching and Teacher Education*, 33, 34–44. <https://doi.org/10.1016/j.tate.2013.02.004>
- Matsuyama, Y. (2020). Hierarchical linear modeling (HLM). In M. D. Gellman (Ed.), *Encyclopedia of Behavioral Medicine*. Cham: Springer. https://doi.org/10.1007/978-3-030-39903-0_407
- MoEYS. (2008). *Policy on education for children with disabilities*. Cambodia: Phnom Penh.
- MoEYS. (2017a). *Curriculum framework for bachelor of arts (education) at teacher education college (TEC)*. Cambodia: Phnom Penh.
- MoEYS. (2017b). *Curriculum framework for primary school teacher (12+2) with credit system*. Cambodia: Phnom Penh.
- MoEYS. (2017c). *Teacher training guide in inclusive education for teacher trainers*. Cambodia: Phnom Penh.
- MoEYS. (2018). *Policy on inclusive education*. Cambodia: Phnom Penh.
- MoEYS. (2020). *Perspective on the current status of education policy implementation*. Cambodia: Phnom Penh.
- MoEYS. (2022a). *Education congress: The education, youth, and sport performance in the academic year 2020-2021 and goals for the academic year 2021-2022*. Cambodia: Phnom Penh.
- MoEYS. (2022b). *Public education statistics and indicators 2021-2022*. Cambodia: Phnom Penh.
- MoEYS. (2023). *Statistics on students with disabilities at pre-primary, primary, lower secondary, and upper secondary schools in Cambodia in the academic year 2021-2022*. Cambodia: Phnom Penh.
- Monsen, J. J., Ewing, I. L., & Kwoka, M. (2014). Teachers' attitudes towards inclusion, perceived adequacy of support and classroom learning environment. *Learning Environments Research*, 17, 113–126. <https://doi.org/10.1007/s10984-013-9144-8>
- Monteiro, E., Kuok, A. C., Correia, A. M., Forlin, C., & Teixeira, V. (2019). Perceived efficacy of teachers in Macao and their alacrity to engage with inclusive education. *International Journal of Inclusive Education*, 23(1), 93–108. <https://doi.org/10.1080/13603116.2018.1514762>
- MoP. (2020). *General population census of the kingdom of Cambodia 2019: National report on final census results*. Cambodia: Phnom Penh.
- Musyoka, M. M., & Diane Clark, M. (2017). Teachers' perceptions of individualised education programme (IEP) goals and related services. *Journal of Developmental and Physical Disabilities*, 29, 5–23. <https://doi.org/10.1007/s10882-015-9462-0>
- Nishio, A. (2019). History and current situation of education for children with disabilities in Cambodia: A gray literature review. *Scientific Research*, 10, 1753–1763. <https://doi.org/10.4236/ce.2019.108125>
- OECD. (2016). *PISA 2015 results: Policies and practices for successful schools*.
- Peebles, J. L., & Mendaglio, S. (2014). The impact of direct experience on preservice teachers' self-efficacy for teaching in inclusive classrooms. *International Journal of Inclusive Education*, 18(12), 1321–1336. <https://doi.org/10.1080/13603116.2014.899635>
- Pijl, S. J. (2010). Preparing teachers for inclusive education: Some reflections from the Netherlands. *Journal of Research in Special Educational Needs*, 10, 197–201. <https://doi.org/10.1111/j.1471-3802.2010.01165.x>
- Polat, F. (2011). Inclusion in education: A step towards social justice. *International Journal of Educational Development*, 31(1), 50–58. <https://doi.org/10.1016/j.ijedudev.2010.06.009>
- Raudenbush, S.W., Bryk, A.S., Cheong, Y.F., Congdon, R.T., & Toit, M.D. (2019). *HLM 8: Hierarchical linear and nonlinear modeling*. Scientific Software International.
- Ravet, J., & Mtiika, P. (2021). Educational inclusion in resource-constrained contexts: A study of rural primary schools in Cambodia. *International Journal of Inclusive Education*, 1–22. <https://doi.org/10.1080/13603116.2021.1916104>
- Round, P. N., Subban, P. K., & Sharma, U. (2016). I don't have time to be this busy.' Exploring the concerns of secondary school teachers towards inclusive education. *International Journal of Inclusive Education*, 20(2), 185–198. <https://doi.org/10.1080/13603116.2015.1079271>
- Shah, R., Das, A., Desai, I., & Tiwari, A. (2016). Teachers' concerns about inclusive education in Ahmedabad, India. *Journal of Research in Special Educational Needs*, 16(1), 34–45. <https://doi.org/10.1111/1471-3802.12054>
- Sharma, U., Forlin, C., & Loreman, T. (2008). Impact of training on pre-service teachers' attitudes and concerns about inclusive education and sentiments about persons with disabilities. *Disability Society*, 23(7), 773–785. <https://doi.org/10.1080/09687590802469271>
- Sharma, U., Loreman, T., & Forlin, C. (2012). Measuring teacher efficacy to implement inclusive practices. *Journal of Research in Special Educational Needs*, 12(1), 12–21. <https://doi.org/10.1111/j.1471-3802.2011.01200.x>
- Sharma, U., & Sokal, L. (2015). The impact of a teacher education course on pre-service teachers' beliefs about inclusion: An international comparison. *Journal of Research in Special Educational Needs*, 15(4), 276–284. <https://doi.org/10.1111/1471-3802.12043>
- Šiška, J., & Suchánek, O. (2015). Access to education for learners with disabilities in Cambodia as a post-conflict country. *Development, Environment and Foresight*, 1(2), 77–91.
- Specht, J., McGhie-Richmond, D., Loreman, T., Mirenda, P., Bennett, S., Gallagher, T., & Cloutier, S. (2016). Teaching in inclusive classrooms: Efficacy and beliefs of Canadian preservice teachers. *International Journal of Inclusive Education*, 20(1), 1–15. <https://doi.org/10.1080/13603116.2015.1059501>
- Specht, J. A., & Metsala, J. L. (2018). Predictors of teacher efficacy for inclusive practice in pre-service teachers. *Exceptionality Education International*, 28(3), 67–82. <https://doi.org/10.5206/eei.v28i3.7772>
- Sullivan, L. M., Dukes, K. A., & Losina, E. (1999). An introduction to hierarchical linear modelling. *Statistics in Medicine*, 18(7), 855–888. [https://doi.org/10.1002/\(SICI\)1097-0258\(19990415\)18:7<855::AID-SIM117>3.0.CO;2-7](https://doi.org/10.1002/(SICI)1097-0258(19990415)18:7<855::AID-SIM117>3.0.CO;2-7)
- Tsakiridou, H., & Polyzopoulou, K. (2014). Greek teachers' attitudes toward the inclusion of students with special educational needs. *American Journal of Educational Research*, 2(4), 208–218. <https://doi.org/10.12691/education-2-4-6>
- Tschannen-Moran, M., Hoy, A. W., & Hoy, W. K. (1998). Teacher efficacy: Its meaning and measure. *Review of Educational Research*, 68(2), 202–248. <https://doi.org/10.3102/00346543068002202>
- UNESCO. (1994). *The Salamanca statement and framework for action on special needs education*. Spain: Salamanca.
- UNESCO. (2005). *Guidelines for inclusion: Ensuring access to education for all*. France: Paris.
- UNESCO. (2015). *Incheon declaration and framework for action for the implementation of sustainable development goal 4*. South Korea: Incheon.
- UNICEF. (2003). *Inclusive education initiatives for children with disabilities: Lessons from the East Asia and Pacific region*. Thailand: Bangkok.
- Urton, K., Wilbert, J., Krull, J., & Hennemann, T. (2023). Factors explaining teachers' intention to implement inclusive practices in the classroom: Indications based on the theory of planned behaviour. *Teaching and Teacher Education*, 132, Article 104225. <https://doi.org/10.1016/j.tate.2023.104225>
- Valdés, R., & Gómez-Hurtado, I. (2023). Barriers and facilitators for leading the development of inclusive school cultures in Chile. *International Journal of Inclusive Education*, 1–17. <https://doi.org/10.1080/13603116.2023.2241861>
- Van Miegheem, A., Struyf, E., & Verschueren, K. (2022). The relevance of sources of support for teachers' self-efficacy beliefs towards students with special educational needs. *European Journal of Special Needs Education*, 37(1), 28–42. <https://doi.org/10.1080/08856257.2020.1829866>
- Vogiatzi, C. A., Charitaki, G., Kourkoutas, E., & Forlin, C. (2022). The teacher efficacy for inclusive practices (TEIP) scale: Further evidence for construct validity in Greek-speaking teachers. *Prospects*, 1–17. <https://doi.org/10.1007/s11125-022-09605-w>
- Weisel, A., & Dror, O. (2006). School climate, sense of efficacy and Israeli teachers' attitudes toward inclusion of students with special needs. *Education, Citizenship and Social Justice*, 1(2), 157–174. <https://doi.org/10.1177/1746197906064677>
- Woltman, H., Feldstain, A., MacKay, J. C., & Rocchi, M. (2012). An introduction to hierarchical linear modelling. *Tutorials in Quantitative Methods for Psychology*, 8(1), 52–69.
- Woodcock, S., Sharma, U., Subban, P., & Hitches, E. (2022). Teacher self-efficacy and inclusive education practices: Rethinking teachers' engagement with inclusive practices. *Teaching and Teacher Education*, 117, Article 103802. <https://doi.org/10.1016/j.tate.2022.103802>
- Wray, E., Sharma, U., & Subban, P. (2022). Factors influencing teacher self-efficacy for inclusive education: A systematic literature review. *Teaching and Teacher Education*, 117, Article 103800. <https://doi.org/10.1016/j.tate.2022.103800>
- Yada, A., & Alnahdi, G. H. (2021). A comparative study on Saudi and Japanese in-service teachers' attitudes towards inclusive education and self-efficacy in inclusive practices. *Educational Studies*, 1–19. <https://doi.org/10.1080/03055698.2021.1969646>
- Yada, A., & Savolainen, H. (2017). Japanese in-service teachers' attitudes toward inclusive education and self-efficacy for inclusive practices. *Teaching and Teacher Education*, 64, 222–229. <https://doi.org/10.1016/j.tate.2017.02.005>
- Yada, A., & Savolainen, H. (2019). Japanese and Finnish teachers' perceptions and self-efficacy in inclusive education. *Journal of Research in Special Educational Needs*, 19, 60–72. <https://doi.org/10.1111/1471-3802.12478>
- Yan, Z., & Sin, K. F. (2014). Inclusive education: Teachers' intentions and behaviour analysed from the viewpoint of the theory of planned behaviour. *International Journal of Inclusive Education*, 18(1), 72–85. <https://doi.org/10.1080/13603116.2012.757811>